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Abstract

The objective of this project is to numerically solve the equations of fluid motion with heat transfer, i.e., the Navier-Stokes equations coupled with the energy conservation equation, in a multidimensional manner (CFD & HT methods). The methodology will be based on finite-volume techniques, using the fractional-step method to solve the coupling between the pressure and velocity fields, i.e., the conservation of mass and momentum equations. In a further step, the energy conservation equation will be incorporated, enabling the study of convection heat transfer problems. The analysis will begin with studies of conduction heat transfer problems (i.e., 4-materials problem). In a second step, a given velocity field will be incorporated to study the numerical diffusion problem in the solution of the convection-diffusion equation (i.e., Smith-Hutton problem). In the third step, a fluid motion solver for laminar flows will be developed, i.e., the mass and momentum conservation equations; finally, the energy conservation equation will be added to the fluid solver. The computational codes will be verified through grid-refinement studies in both space and time, using corresponding benchmark cases at each step. The investigation of the selected problems will include parametric studies considering variations in geometry, thermophysical properties, and boundary conditions.

Introduction

Engineering as a practice has always centered around finding optimized solutions for real-world applications. Throughout history, countless names have devoted their efforts towards developing mathematical tools and models that allowed the next generation of engineers to further expand the knowledge over physical phenomena. Paired with the rapid increase in computing capacity seen over the last few decades, there is a duty to continue their work with the new tools available in our time. In the field of Computer Fluid Dynamics and Heat Transfer (CFD & HT), the resolution of the Navier-Stokes equations presents an important challenge that can be optimized for a wide variety of industrial applications. The Navier-Stokes (NS) equations describe the motion of viscous fluids and the complex interactions involved in the phenomena by evaluating the conservation of mass, momentum and energy. However, NS equations are non-linear since the velocity and pressure equations (velocity & mass) in the system are mutually dependent. This adds a layer of complexity for solving the system of equations and in consequence there is no general solution that can be directly solved. However, all conservation equations follow the same physical principle that describes transport phenomena.

This report will cover the development of numerical tools for the resolution of Convection-Diffusion problems while consolidating the knowledge related to the transport phenomena and the application of numerical methods for their resolution. The spatial discretization in this project will follow a finite volume method to discretize the analyzed domain into a finite amount of control volumes. The model will be developed in stages of increasing complexity, starting with pure diffusion, convection-diffusion, and finally coupled NS problems; using well-known benchmarks to validate the obtained results at each stage. During the development, the effect of the different numerical parameters involved in the model will be studied to further understand the mathematical implications of using a numerical tools like the one proposed with this project. Furthermore, the nature of each different case will require the implementation of different numerical solver algorithms that address the optimization problem in different manners.

Finite Volume Method (FVM)

As mentioned above, the model will separate the domain into rectangular control volumes, which will then be evaluated independently considering all their interactions with the adjacent nodes or boundaries. Figure 1 shows the representation of an interior node surrounded by adjacent nodes on all sides. As it can be seen, the four neighbours will be assigned a letter corresponding to their cardinal direction, using capital letters to refer to values located at the adjacent node centroid, and uncapitalized letters to refer to values located at the interface between the main node and the respective adjacent node. This presents some particular difficulties, as values are only known at the node centroids, and different schemes for interpolation may be used to estimate the values at the interface for certain models.

Governing Transport Equations

When studying transport phenomena, there are two main mechanisms that govern the propagation of a transport variable ϕ . These are known as **Convection** and **Diffusion**, and classify the phenomena depending on what causes the transport of the transport variable ϕ . Diffusion refers to transport caused by molecular movement within the body, while Convection refers to transport caused by bulk movement of the body. The governing transport equation for this

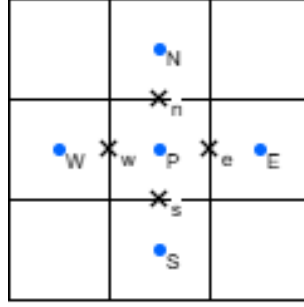


Figure 1: Node distribution for a single control volume P . Subindexes for each adjacent node correspond to their cardinal direction relative to the main node. Using the capitalized letter to represent values located at the adjacent node centroid (W , E , S , N) and the uncapitalized alternative for values located at the wall corresponding to the interaction between the main node and the adjacent node.

phenomena is given by Eq. (1), which is separated into four terms relating to the physical variation of ϕ . The accumulation term describes the magnitude variation over time, the convective term describes the variation due to convection, the diffusive term describes the variation due to diffusion, and the source term describes any additional source contributions.

$$\underbrace{\frac{\partial(\rho\phi)}{\partial t}}_{\text{Accumulation}} + \underbrace{\nabla(\rho v\phi)}_{\text{Convective}} = \underbrace{\nabla(\Gamma\nabla\phi)}_{\text{Diffusive}} + \underbrace{S_P}_{\text{Source}} \quad (1)$$

The continuous governing equation needs to be integrated over a control volume V_P with a defined time-step Δt . By applying the Gauss divergence theorem shown in Eq. 2, the convective and diffusive terms can be converted into surface integrals, which will evaluate the flow through the cell interface.

$$\int_{V_P} \frac{\partial(\rho\phi)}{\partial t} dV = \oint_{\partial V_P} \rho \mathbf{v}\phi \cdot \hat{n} dA = \oint_{\partial V_P} \Gamma \nabla\phi \cdot \hat{n} dA + \int_{V_P} S_P dV \quad (2)$$

Each surface integral can then be approximated as a discrete sum over the four cell faces (west, east, south, north). The resulting discretized forms are shown by Eqs. (3)–(5)

$$\frac{\partial(\rho\phi)}{\partial t} = \rho \frac{\phi_P - \phi_P^0}{\Delta t} \quad (3)$$

The accumulation term follows a first-order implicit approximation, where ϕ_P^0 corresponds to the value of node P at the current time-step and ϕ_P corresponds to the value at the next time-step.

$$\oint_{\partial V_P} \Gamma \nabla\phi \cdot \hat{n} dA = \Gamma_w \frac{\phi_W - \phi_P}{d_{PW}} S_w - \Gamma_e \frac{\phi_P - \phi_E}{d_{PE}} S_e + \Gamma_s \frac{\phi_S - \phi_P}{d_{PS}} S_s - \Gamma_n \frac{\phi_P - \phi_N}{d_{PN}} S_n \quad (4)$$

The diffusive term approximates the gradient at the face by interpolating the values from the adjacent nodes, multiplied by the diffusivity coefficient Γ and the surface at the interface S_k , and dividing by the distance between the central node and one of the adjacent nodes d_{PK} .

$$\oint_{\partial V_P} \rho \mathbf{v} \phi \cdot \hat{n} dA = \rho(v_e \phi_E S_e - v_w \phi_W S_w + v_n \phi_N S_n - v_s \phi_S S_s) \quad (5)$$

The convective term uses the velocity components located at the cell wall faces, considered in the expression with their corresponding geometrical surface value. For this term, the convective scheme can considerably change the behaviour of the model.

By substituting all previously mentioned equations and organizing the different terms, the expression can be reorganized to depend only on the magnitude of the current node P and its neighbours. Eq. (6) shows the standard FVM form for an interior cell with 4 adjacent nodes.

$$a_P \phi_P = a_W \phi_W + a_E \phi_E + a_S \phi_S + a_N \phi_N + b_P \quad (6)$$

The coefficients a_W, a_E, a_S, a_N include the effect of both the diffusive and convective phenomena. When using this discretization, the coefficient for the current cell $a_P = a_W + a_E + a_S + a_N + a_P^0$ will respect diagonal dominance due to the presence of the accumulation term $a_P^0 = \rho V_P / \Delta t$. This equation will be iteratively solved for every node in the discretized domain at each time-step. Each chapter will include a brief explanation of the specific expressions used for each coefficient for their respective application.

Motivation

Objectives

1. Develop a modular FVM solver using C++ capable of accurately simulating transport phenomena of increasing complexity. Using several benchmark cases to validate the accuracy of the implementation at each stage of the model, and testing the performance and behaviour of the different numerical methods implemented in the model once validated.
2. Expand the knowledge about numerical methods applied to CFD & HT and use it to solve and understand the resolution methodology of complex transport phenomena. Simultaneously using the opportunity to further develop a consistent and organized code structure and learning more about C++ and its capabilities as a programming language.

Scope

The model developed in this project will solve three main models, testing a variety of benchmark problems for each one of them.

1. The **pure diffusion** model will evaluate a purely diffusive heat conduction test scenario in 1D and 2D. This first model has the simplest discretization and will serve as a base implementation to test the data structure design for the code implementation. It will also serve as a first introduction to numerical parameters such as the time-step and temporal scheme, and allow for testing to be done regarding their effect on the performance of the

model. For the 1D case, the code will be validated against the analytical solution for all implemented boundary combinations in a **1D Rod Case**. For the 2D case, the model will be validated against the **4 Materials Case** benchmark published by PENDING REFERENCE [?]. In addition, this first case will be used to compare the Gauss-Seidel (GS) and Conjugate Gradient (CG) numerical algorithms in terms of performance.

2. The **convection-diffusion** model will evaluate a complete convection-diffusion in 2D showing the spatial evolution of a transport variable for a fixed flow field. This model completes the transport equation and also introduces the discretization model proposed by Patankar [1], which implements a multi-scheme model for the convective scheme. The model will be validated against the **Smith-Hutton Case** benchmark published by PENDING REFERENCE [?]. This section will also require the implementation of a new BiConjugate Gradient Stabilized (BiCGSTAB) numerical solver algorithm to solve the complexities introduced by the convective term.
3. The **Navier-Stokes** model will evaluate a coupled 2D Navier-Stokes solver. This model will introduce the concepts of Fractional Step Method (FSM) and Staggered Meshes, requiring a severe overhaul for the code structure. The NS equation model can be simplified in multiple ways, depending on the physical phenomena trying to be described. As such, two different test scenarios of increasing complexity will be evaluated. The first one consists of the **Lid Driven Cavity Case** benchmark published by PENDING REFERENCE [?], which simplifies the model to include only the mass conservation component and the 2 momentum conservation components. The second case consists of the **Differentially Heated Cavity Case** benchmark published by PENDING REFERENCE [?], which introduces an additional energy conservation component. Fractional Step Method (FSM) and Staggered Meshes will also be introduced and their effect in model convergence will be studied.

1 Pure Diffusion Solver

This chapter will explain the implementation of a pure diffusion model, which is the simplest version of a transport equation introduced in Chapter . This scenario is obtained by setting the velocity field to zero ($\mathbf{v} = 0$), which eliminates the convective term from the expression. This case describes the physical phenomena of heat conduction through solid bodies, where energy is transmitted exclusively through diffusion as described by Fourier's Law, shown in Eq. (7). The simplicity of the model lends itself to be a good introduction for studying the numerical discretization and validating the core infrastructure of the code implementation. The core components **Material**, **Mesh**, **Discretizer**, **Solver**, and **Probe** were defined on this stage and will remain as the fundamental structure of the model moving forward.

$$q_{cond} = -\lambda \nabla T \quad (7)$$

Following the previously mentioned assumption of a stationary flow field, the convective term can be removed from the governing transport equation, reducing it to the expression shown in Eq. 8. For a heat conduction phenomena, the diffusivity Γ corresponds to the thermal conduction coefficient λ , while the source term S_P corresponds to the internal heat generation q_V . As such, the expanded transport equation becomes Fourier's heat equation, shown in Eq. (8).

$$\rho c_P \frac{\partial T}{\partial t} = \nabla(\lambda \nabla T) + q_V \quad (8)$$

By applying the FVM described in Chapter and integrating over a control volume V_P , the following discretized form of the equation is shown in Eq. (9).

$$\rho c_P V_P \frac{T_P - T_P^0}{\Delta t} = \lambda_w S_w \frac{T_W - T_P}{d_{PW}} + \lambda_e S_e \frac{T_E - T_P}{d_{PE}} + \lambda_s S_s \frac{T_S - T_P}{d_{PS}} + \lambda_n S_n \frac{T_N - T_P}{d_{PN}} + q_V V_P \quad (9)$$

In addition, an important observation regarding pure-diffusion is that it requires the use of the harmonic to calculate the diffusion coefficient when the cell wall is located at a material interface. The expression used to obtain the harmonic mean between two adjacent nodes with different properties is shown in Eq. (10).

$$\lambda_k = \frac{2\lambda_P \lambda_K}{\lambda_P + \lambda_K} \quad (10)$$

By expanding Eq. (9) and reorganizing the terms the same expression shown in Eq. (6) can be obtained for this particular problem. Table 1 shows a summary of the discretized coefficient expressions

1.1 Boundary Conditions

Since boundary conditions are imposed instead of corresponding to the interaction of the center node with a neighboring node, their implementation in the code needs a special formulation.

Coefficient	Expression
a_W	$\frac{\lambda_w S_w}{d_P W}$
a_E	$\frac{\lambda_e S_e}{d_P E}$
a_S	$\frac{\lambda_s S_s}{d_P S}$
a_N	$\frac{\lambda_n S_n}{d_P N}$
b_P	$a_P^0 T_P^0 + q_V V_P$

Table 1: Summary of discretized coefficients for standard FVM formulation for an interior cell. Where a_k corresponds to the coefficient describing the interaction with an adjacent node, and b_P is the right-hand side (RHS) term that describes the influence of the previous step and the source.

The current model supports 3 different types of boundary conditions; namely **Dirichlet**, **Neumann**, and **Robin**. The explanation concerning the implementation of boundary conditions will be done taking a west boundary as reference; the same concept detailed in this scenario will apply for the other boundaries with their respective coefficients.

Dirichlet boundary conditions directly assign the variable value at the boundary. As such, the gradient can be approximated by using the distance between the cell centroid and the boundary wall d_{Pw} and following the same discretization definitions shown in Table 1. The coefficient term will be added to the diagonal coefficient a_P and the assigned temperature is imposed by introducing it as part of b_P .

$$a_W = 0 \quad a_P += D_w = \frac{\lambda_w S_w}{d_{Pw}} \quad b_P += D_w T_{BC} \quad (11)$$

Neumann boundary conditions assign the flux at the surface rather than the magnitude. As such, the flux value is already known at the surface and can be directly included into the RHS term of the discretization. No additional terms need to be added to the diagonal coefficient.

$$a_W = 0 \quad b_P += q_{BC} S_w \quad (12)$$

Robin boundary conditions, also known as convection / hybrid boundary conditions, assigns a variable heat flux that relates the temperature at the node T_P with the temperature of a far-away fluid T_∞ . The added conductance needs to be included in the diagonal term a_P and to the RHS part of the equation.

$$a_W = 0 \quad a_P += \alpha S_w \quad b_P += \alpha S_w T_\infty \quad (13)$$

Implementation and validation of the 1D and 2D cases are discussed in the following subsections.

1.2 1D Transient Conduction - 1D Rod Case

In this section, the implementation of the heat conduction model for the **1D Rod Case** will be developed. The problem consists of a thin $1m$ bar with adiabatic walls on the south/north faces and Dirichlet boundary conditions for the west/east faces. The west wall is set at $T_w = 100^\circ C$, while the eastern wall is set at $T_e = 20^\circ C$, with initial conditions set at $T_0 = 30^\circ C$. Furthermore, the rod is defined as being made of copper with properties $\lambda = 400W/(m \cdot K)$, $\rho = 8960kg/m^3$, and $c_P = 380J/(kg \cdot K)$. Figure 2



Figure 2: Diagram of 1D Rod Case, including original boundary conditions proposed for that case [?].

1.2.1 Model Implementation

The model implementation was designed to work using a configuration file that includes all the relevant parameters required to define a simulation. This ensures that the model will be robust enough to handle a wide range of simulations without needing any modifications. A summary of all the different parameters included in the file is shown in Table 2. A detailed explanation of each one of the groups and their relevant information is shown below. **Physical** includes the initial temperature and a list of the available materials and their thermophysical properties. **Geometrical** includes the geometric parameters; the model can only support rectangular shapes in this implementation and changes will be made when more complex geometries are studied. Sections must be defined with a material, the number of nodes for that specific section N_s , beginning and ending coordinates x_0 and x_1 , and internal heat generation q_V . **Numerical** includes the numerical parameters; the model currently has both the Gauss-Seidel (GS) and Conjugate Gradient (CG) solvers implemented and both will be compared as a first step before performing the relevant studies discussed ahead. **Mesh** includes the option to choose from four different algorithms, as well as all relevant parameters for each of these alternatives. **Boundary Conditions** includes an array describing the boundaries for the model and can currently support Dirichlet, Neumann, and Convection boundary conditions. In addition, it supports variable temperatures for Dirichlet boundary conditions.

1.2.2 Temporal Scheme

One of the first numerical parameters to be evaluated in this project is the temporal scheme. This parameter determines the relevance that each time-step will have in the calculation of the next one. The **explicit** scheme, relies exclusively on information available from the current time-step, described as a function $f(T^{n-1})$ and requires a small enough time-step to ensure stability, given by Eq. (14). On the other hand, the **implicit** scheme relies exclusively on information from the

Group	Parameter	Description
Physical Data	T0 materials	Initial value for temperature map. List of vectors with thermophysical properties corresponding to each material.
Geometrical Data	width height sections	Geometric dimension. Geometric dimension. List of vectors indicating the distribution of materials in the simulation. Also includes amount of nodes and internal heat generation.
Numerical Data	scheme solver maxIterations endTime timeStep tolNumeric tolTemporal	Scheme for simulation. Solver algorithm for simulation. Limit of iterations per timestep. Last instant of the simulation. Time increment after each instant. Tolerance for numerical solver. Tolerance for steady-state convergence.
Mesh Data	meshAlgorithm strength centering kappa delta	Algorithm for mesh generation. Cubic algebraic mesh parameter. Cubic algebraic mesh parameter. Power law mesh parameter. Hyperbolic tangent mesh parameter.
Boundary Data	boundaries	List of vectors describing the boundary conditions on each side.

Table 2: Configuration file structure for 1D Rod Case. Showing the design structure for the metadata used to define the simulations.

time-step being currently calculated, being a function of $f(T^n)$. Finally, the **Crank-Nicolson** scheme employs a second-order scheme that averages the estimation made by both the explicit and implicit temporal schemes, being a function of $f(0.5(T^{n-1} + T^n))$.

$$\Delta t_{max} = \frac{\Delta x^2}{2\alpha} \quad (14)$$

1.2.3 Solver Comparison

An additional decision to make when solving this kind of problems is the numerical solver to be used during simulations. While the matrix definition remains constant in both cases, as both require a sparse matrix in the form $AT = b$, selecting the right solver can have a big impact in the performance of the solver. In this project, an initial comparison between the Gauss-Seidel (GS) and Conjugate Gradient (CG) algorithms will be made with the objective of evaluating performance and efficiency. A brief introduction to both algorithms is shown below before showing the results of said comparison.

Gauss-Seidel The GS method solves linear systems by iteratively cycling through each node in sequence and updating them by using the most recent values from its adjacent nodes. Each cell needs to be calculated directly by using Eq. (15), which delays the propagation of information from one cell to the others. While convergence is a guarantee when the matrix is diagonally

dominant, it is really susceptible to mesh size as it will directly impact the number of operations.

$$T_P^{n+1} = \frac{1}{a_P} (b_P - \sum_k a_k T_k^*) \quad (15)$$

Conjugate Gradient The CG method is an iterative Krylov subspace solver designed specifically for Symmetric Positive Definite (SPD) systems. For heat conduction problems, this matches the diffusion coefficient matrix since the interaction between two adjacent nodes is identical in both directions (symmetrical) and is diagonally dominant due to the accumulation term in a_P (positive definite). Contrary to the GS method, rather than updating the values one by one, CG minimizes the residual over the sequence of search directions $\mathbf{p}_k$ that are mutually A -conjugate:

$$\mathbf{p}_i^T A \mathbf{p}_j = 0, \quad i \neq j \quad (16)$$

This guarantees that no iteration undoes the progress made by previous ones. Because of this capacity of propagating information, CG converges in maximum N iterations, being N the amount of unknowns. It is also important to highlight that CG is only valid for SPD systems. As it will be explained in Chapter 2, once symmetry is broken due to the convective term, the solver algorithm needs to be changed to a more generalized variant to be able to handle non-SPD scenarios.

Results These two metrics will be used to evaluate the performance of the solvers. Only one will be used to perform the numerical parameter studies shown below; as such, these metrics will be used to compare both solutions. The base simulation used in this study will be the **1D Rod Case** described by PENDING CITATION [?], with the addition of an internal heat source for a better comparison with the steady-state analytical solution. This simulation will use $N = 100$, $\Delta t = 0.5$ and a Crank-Nicolson scheme with a uniform mesh. The results for both cases are shown in Figure 3. In both metrics, it can be seen that CG outperforms GS by a considerable margin; converging faster and obtaining lower residue. As mentioned above, the CG algorithm was designed specifically for SPD coefficient matrices, which is the case for a pure-diffusion model. As described above, CG computes the conjugated search directions of all elements in the matrix, which guarantees convergence in at most N steps. On the other hand, GS updates one node at a time using the most recent values and propagates information one cell per iteration, making it slower than CG. From these results, the decision to use CG for the numerical solver was made.

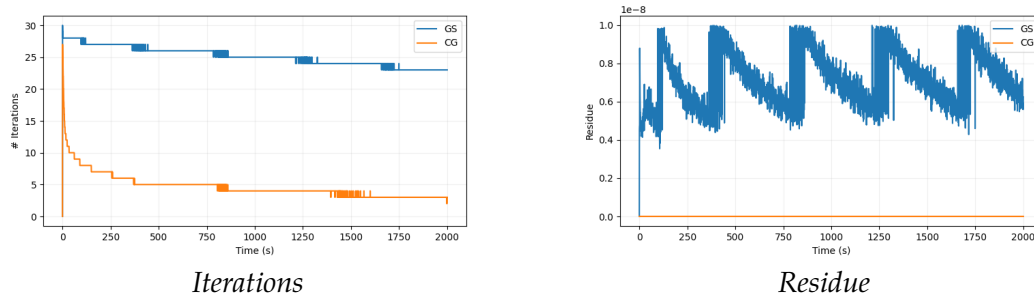


Figure 3: Performance comparison between Gauss-Seidel and Conjugate Gradient solvers. Number of iterations until solution (left) and solver residue after convergence (right).

1.2.4 Numerical Parameters Study

For this test, the time step Δt and the number of nodes N are modified to analyze the number of iterations and compare them with the analytical solution. Both parameters will be independently swept across their respective range of values, while keeping the other parameter at a constant value. A summary of the values for each one of these parameters is shown in Table 3, highlighting the values used for the base case.

Parameter	Values
Δt	0.1, 0.5 , 1.0, 2.0, 5.0
N	50, 100 , 200, 400

Table 3: Detail of numerical parameters to be studied. Each variable will remain at their respective highlighted value while the other one sweeps through their range.

Δt Sweep The results obtained from the tests explained above are shown in Figure 4. As expected, the number of iterations remains unchanged for the explicit scheme, as it is meant to be a direct calculation. On the other hand, both the Crank-Nicolson and implicit schemes require more iterations per time step as it increases. This is mainly due to the accumulation term decreasing in magnitude as Δt increases, while the adjacent node coefficients do not decrease with Δt . This in turn makes the system less diagonally dominant and slower to converge. Compared to the analytical solution, the explicit scheme shows a significant error for $0.1 < \Delta t$, while a small increase could be seen for the Crank-Nicolson and implicit schemes. As a second-order accurate method, Crank-Nicolson shows lower error compared to implicit, which is a first-order accurate method. Overall, the Crank-Nicolson scheme has shown better performance in terms of number of iterations needed and precision compared to the analytical solution. Furthermore, the results for explicit seem to stagnate at a constant value after the lowest time step value tested, suggesting that the limit set by Eq. 14 has been exceeded. The computation time is seen to be similar between the Crank-Nicolson and implicit schemes, with a small advantage for the implicit scheme.

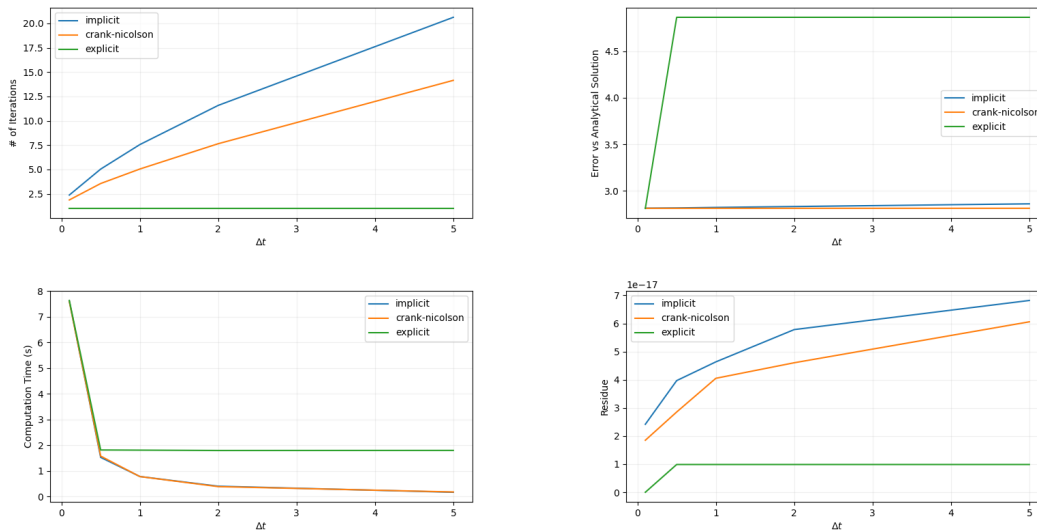


Figure 4: Δt sweep results.

***N* Sweep** The results obtained for the tests explained above are shown in Figure 5. Similarly to the previous case, the explicit scheme is a direct calculation and does not require more iterations. In addition, both Crank-Nicolson and implicit schemes show an increase in the number of iterations for the Crank-Nicolson and implicit schemes, suggesting better performance for Crank-Nicolson as the mesh gets finer. Since a larger mesh means more finite volumes, the amount of calculations performed by the model will increase. Compared to the analytical solution, the explicit scheme still showed higher values when compared to the Crank-Nicolson and implicit schemes; however, it seems to get closer as the number of nodes increases. In terms of computation time, both the Crank-Nicolson and implicit schemes saw a slight increase, while the explicit scheme saw a major difference as the mesh got finer.

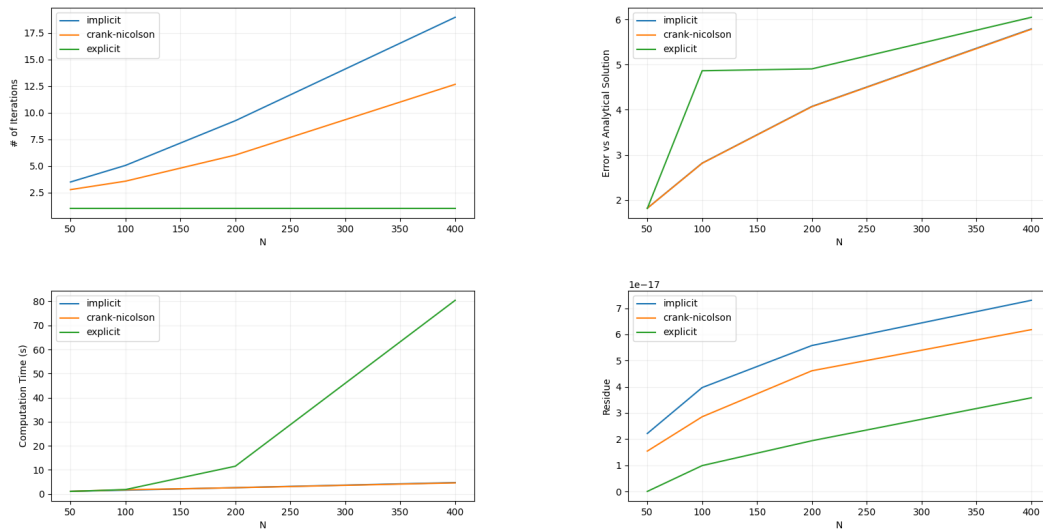


Figure 5: *N* sweep results.

1.2.5 Model Validation

In addition to the Dirichlet boundary condition established in the problem statement, Neumann and Convection boundary conditions have also been implemented in the model. Furthermore, support for variable temperatures for Dirichlet boundary conditions was implemented. To validate this implementation, a comparison will be made with the analytical solution for a steady-state scenario with the transient-state model once it reaches equilibrium. A summary of the pairs of boundary conditions used for the tests is shown in Table 4, while comparisons with the steady state solution are shown in Figure 6. An additional case is included to test the proper implementation of the variable temperature Dirichlet boundary condition. As shown, all pairs evaluated accurately matched the analytical solution. Furthermore, the configuration file can now provide mathematical expressions instead of constant values when defining Dirichlet boundary conditions, as shown by the temperature profile. This capability can now easily be scaled up towards other functions in the model.

$$T(x) = 1 \quad (17)$$

Case	Boundary Conditions
1	Neumann-Dirichlet
2	Convection-Dirichlet
3	Dirichlet-Neumann
4	Dirichlet - Convection
5	$T(t) = 100 + 50 \cos\left(\frac{0.01t}{5\pi}\right)$

Table 4: Cases for model validation. Each case describes a Left-Right pair of boundary conditions to be tested with the 1D Rod Case.

1.3 2D Transient Conduction - 4 Materials Case

In this section, the implementation of a heat conduction model for the **4 Materials Case** will be developed. The problem consists of the rectangular domain shown in Fig. 7, which is composed by 4 different materials with their thermophysical properties being listed in Table 5. Furthermore, the domain has an initial value of $T_0 = 8^\circ C$ and is bound by the boundary conditions shown in Table 6.

Material	ρ [kg/m ³]	c_P [J/(kg · K)]	λ [W/(mK)]
M1	1500	750	160
M2	1600	770	140
M3	1900	810	200
M4	2500	930	140

Table 5

Cavity Wall	Boundary Condition
West	Robin: In contact with fluid at $T_f = 33^\circ C$ and $\alpha = 9W/(m^2K)$
East	Dirichlet: Uniform temperature $T(t) = 8 + 0.005t^\circ C$
South	Dirichlet: Uniform temperature $T = 23^\circ C$
North	Neumann: Uniform $\dot{Q}_{flow}$

Table 6: Boundary conditions for the

1.4 Model Implementation

The model developed for the 2D case will follow the same discretization proposed earlier for the 1D case. However, some modifications to the code structure were needed to support the new functionalities. For example, Probe was expanded to support different types of probes with a better control over the region of the domain that is desired. A summary of the modified parameters included in this iteration of the model is shown in Table 7, summarizing the new parameters in the configuration file. As shown, Probes have now been added as a configurable parameter to allow the user to choose which data is being saved. In addition, refinement was modified to be defined independently for each axis and in separate regions to allow the user better control over the mesh. Furthermore, a Medic class was also included to handle diagnostics and validation of energy conservation in the system. This class is also used as the central tool for debugging as it can easily be modified to log any information relevant to the model.

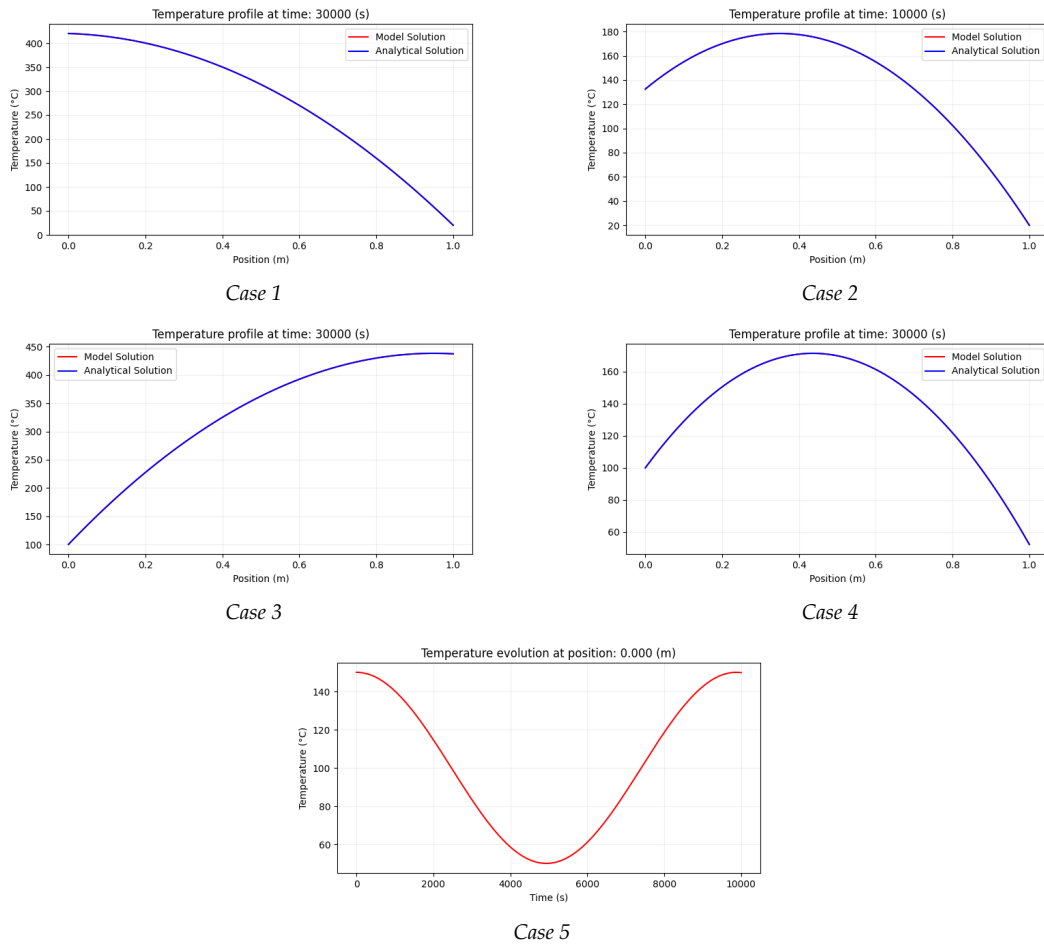


Figure 6: Comparison of model results with analytical solutions and implementation of mathematical expression reading for Dirichlet boundary condition.

1.4.1 Numerical Parameter Study

For this test, timestep Δt and number of nodes N are modified to analyze their impact on the number of iterations and total residue. Both parameters will be independently swept across their respective range of values while keeping the other parameter at a constant value. A summary of the values for each of these parameters is shown in Table 8, highlighting the values used for the base case. In addition, the explicit scheme still requires an imposed limit for the value of Δt ; however the expression for the 2D case changes as described by Eq. 18 due to the presence of simultaneous diffusion in both axis. Contrary to the results shown for the 1D case, there will be no comparison of error with the analytical solution or computation time. This numerical parameter study will focus on comparing the number of iterations and total residue.

$$\Delta t_{max} = \frac{\Delta x^2 \Delta y^2}{2\alpha(\Delta x^2 + \Delta y^2)} \quad (18)$$

Δt Study The results obtained from the tests explained above are shown in Figure 8. As shown, the number of iterations and total residue follow a similar trend to the one seen in the 1D case, increasing with Δt for the implicit and Crank-Nicolson schemes, showing a better perfor-

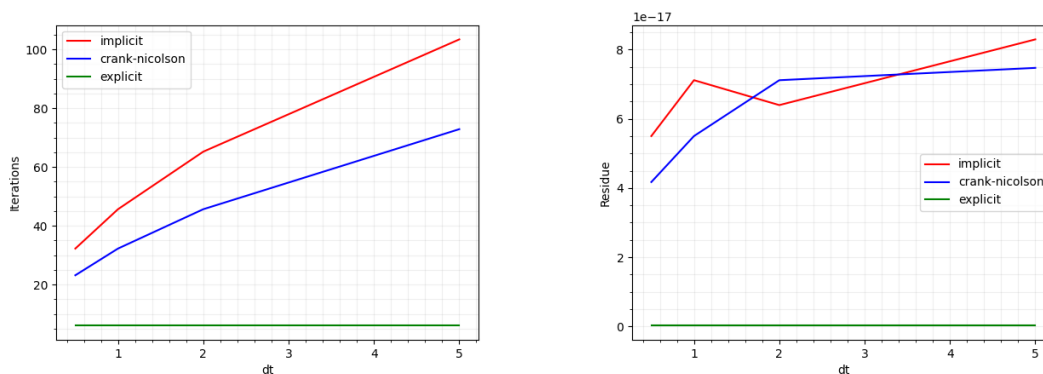


Figure 7: Diagram for 4 Materials Case.

Group	Parameter	Description
Mesh Data	N	List specifying total number of nodes for each axis.
	refinement	List of vectors specifying refinement configuration for each axis.
Probe Data	probes	List of vectors specifying the probes configured for the simulation.

Table 7: Configuration file structure changes, highlighting the expansion in mesh refinement capacity and probe flexibility.

mance for the Crank-Nicolson scheme on both metrics. On the other hand, the explicit scheme did not present any changes during the timestep study. This is due to the timestep being limited by the expression discussed above; which activated for all analyzed cases.

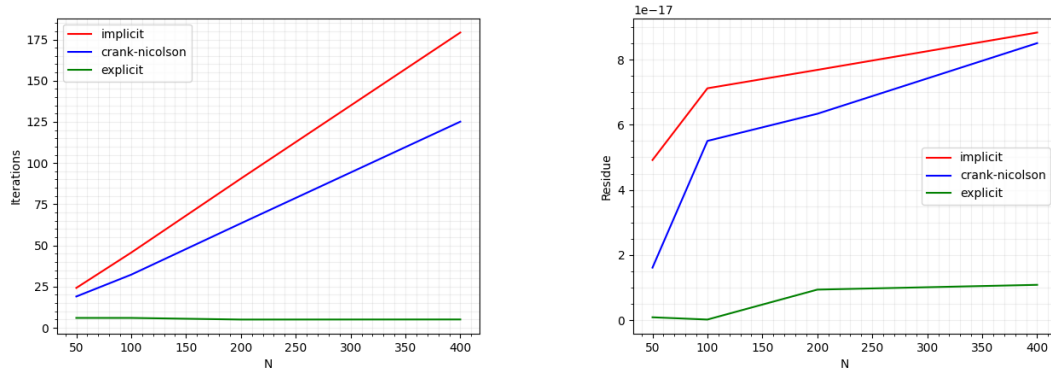
Figure 8: Δt sweep results.

***N* Study** The results obtained from the tests explained above are shown in Figure 9. Just like in the previous test, the presented plots show a similar trend to the one seen in the previous scenario, showing an increase in number of iterations and residue for most cases. Furthermore, the implicit scheme showed higher sensitivity to increases in N compared to the Crank-Nicolson and explicit schemes; with the explicit scheme showing no change at all. On the other hand, while residue was significantly higher for implicit on coarser meshes, the difference between the

Parameter	Values
Δt	0.1, 0.5, 1.0 , 2.0, 5.0
N	50, 100 , 200, 400

Table 8: Value ranges for numerical parameter study.

implicit and Crank-Nicolson schemes became smaller as N increased; with the explicit scheme showing a slight increase in the beginning but less of an impact as N kept increasing.


 Figure 9: N sweep results.

1.4.2 Model Validation

In the same way as done in Section 1.2, boundary conditions were independently analyzed to validate proper implementation in the model by comparing it to the steady-state analytical solution. To simulate this behaviour, an adiabatic Neumann boundary condition ($\frac{\partial T}{\partial n} = 0$) was applied on the faces of the secondary axis in each test. On the other hand, the boundary conditions for the primary axis in the test were configured following the same testing plan described in Section 1.2. The results obtained in this validation successfully replicated the expected behaviour and produced the same plots as those seen in Figure 6 for each axis. Furthermore, a comparison with the *Exercise 3: 4 Materials Case* will be used as a benchmark to complete the validation of the model.

4 Materials Case The results obtained for *Exercise 3: 4 Materials Case* are shown in Fig. 10. As mentioned above, this is a transient simulation and there is no analytical solution available to compare the results with. Instead, this case provides reference values for two separate points at different instants of the simulation. As seen in the results, the model was able to match the values provided for points [0.65, 0.56] and [0.74, 0.72]; as well as showing a temperature distribution consistent with the provided time instants.

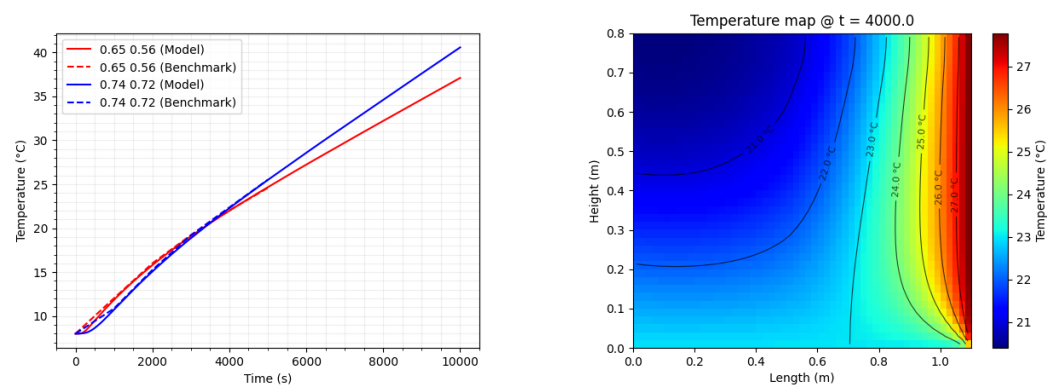


Figure 10: Comparison of model results with results provided in problem statement.

2 Convection-Diffusion Model

This chapter will explain the implementation of a convection-diffusion model. In this case, the complete transport equation shown in Chapter needs to be implemented, considering both the diffusive and convective elements. The model implementation will be validated by solving the Smith-Hutton case and comparing the results to available reference values. In addition, the implementation described in this chapter will follow the discretization model described by Patankar [1], which proposes a generalized formulation that considers the various convective schemes available for numerical resolution. Based on said formulation, the discretized coefficients for a 2D model are shown below in Eqs. 19, 20, 21, 22, 23, 24.

$$a_W = D_w A(|P_w|) + ||F_w, 0|| \quad (19)$$

$$a_E = D_e A(|P_e|) + ||-F_e, 0|| \quad (20)$$

$$a_S = D_s A(|P_s|) + ||F_s, 0|| \quad (21)$$

$$a_N = D_n A(|P_n|) + ||-F_n, 0|| \quad (22)$$

$$a_P = a_W + a_E + a_S + a_N + a_P^0 \quad (23)$$

$$b_P = S_\phi V_P + a_P^0 \phi_P^0 \quad (24)$$

$$a_P^0 = \frac{\rho_P^0 V_P}{\Delta t}$$

As mentioned above, this discretization model includes the additional $A(|P|)$ term, which corresponds to a different function depending on the convective scheme. A summary of the expressions used for the convective schemes implemented in the model is shown in Table 9.

Scheme	$A(P)$
CDS	$1 - 0.5 P $
UDS	1
Hybrid	$ 0, 1 - 0.5 P $
Power Law	$ 0, (1 - 0.1 P)^5 $
Exponential	$\frac{ P }{e^{ P }-1}$

Table 9: The function $A(|P|)$ for different schemes [1].

2.1 2D Convection

This section will cover the implementation of a C++ code for the resolution of a two-dimensional transient-state convection-diffusion transport model. The model works using a configuration file that follows the same structure discussed in the previous chapter with some additional modifications implemented when implementing this section of the model. A summary of the parameters included in the file is shown in Table 10. As it can be seen, the main modifications were the addition of the velocity field, separation of temporal and convective scheme parameters, and the inclusion of a Medic data section designed to control the diagnostics tool. Furthermore, due to the asymmetrical nature of convective transport, an additional numerical solver using the Biconjugate Gradient Stabilized Method had to be implemented to ensure convergence and stability. Finally, the model has fundamentally changed from having defined nodes located at the boundaries to having ghost nodes. In consequence, a new Boundary probe type was introduced to store the values at the boundaries.

Group	Parameter	Description
Numerical Data	tempScheme	Temporal scheme selection for simulation.
	spatScheme	Convective scheme selection for simulation.
	solver	Solver algorithm for simulation.
	tolNumeric	Tolerance for the numerical solver.
	tolTemporal	Tolerance for the steady-state convergence.
	maxIterations	Limit of iterations per timestep.
	endTime	Last instant of the simulation.
Physical Data	timeStep	Time increment after each instant.
	T0	Initial value for temperature map.
	materials	List of vectors with thermophysical properties corresponding to each material.
Geometrical Data	velocityField	List of expressions for each component of the velocity field.
	width	Dimension in the secondary axis.
Mesh Data	sections	List of vectors indicating the distribution of materials in the simulation. Also includes amount of nodes and internal heat generation.
	strength	Cubic algebraic mesh parameter.
	centering	Cubic algebraic mesh parameter.
	kappa	Power law mesh parameter.
	delta	Hyperbolic tangent mesh parameter.
	meshAlgorithm	Algorithm for mesh generation.
	N	List specifying total number of nodes for each axis.
Boundary Data	refinement	List of vectors specifying refinement configuration for each axis.
	boundaries	List of vectors specifying the boundary conditions on each side.
	probes	List of vectors specifying the probes configured for the simulation.
Medic Data	medicOn	Control argument for diagnostic tool.

Table 10: Configuration file structure.

This model will be validated by solving the Smith-Hutton problem, being used both as the test case for the numerical parameter study and for the model validation. The model validation will be done by comparing the model results to reference values provided for cases with $\rho/\Gamma = 10^1, 10^3, 10^6$. The numerical parameter study will compare the number of iterations and total residue for each of these cases for two of the convective schemes implemented for the model; namely Central Difference Scheme (CDS) and Upwind Difference Scheme (UDS).

2.1.1 Numerical Parameter Study

The results obtained for the previously described numerical parameter study are shown in Fig. 11. As shown, using CDS will result in a higher number of iterations as the ratio increases, not being able to reach convergence within the simulation constraints for the case where $\rho/\Gamma = 10^6$. Figure 12 shows the obtained colormap on the last time-step of the simulation. As it can be seen, there is severe checkerboarding across the domain. On the other hand, UDS was able to converge for all values across the tested range. As such, model validation will only show the colormaps for UDS. These divergence is seen due to the Peclet number (Pe), which is a measure of dominance between Convection and Diffusion for a given transport phenomena where $Pe \ll 1$ means the phenomena is diffusion dominant and $Pe \gg 1$ means it is convection dominant. This means that CDS has a particular limitation that can be seen in its definition in Table 9. For simulations where $Pe > 2$, the expression will have an inverted sign and break the diagonal dominance requirement for the linear solver. On the other hand, since UDS only takes values from one side, the coefficients will never see a sign inversion, making it unconditionally stable.

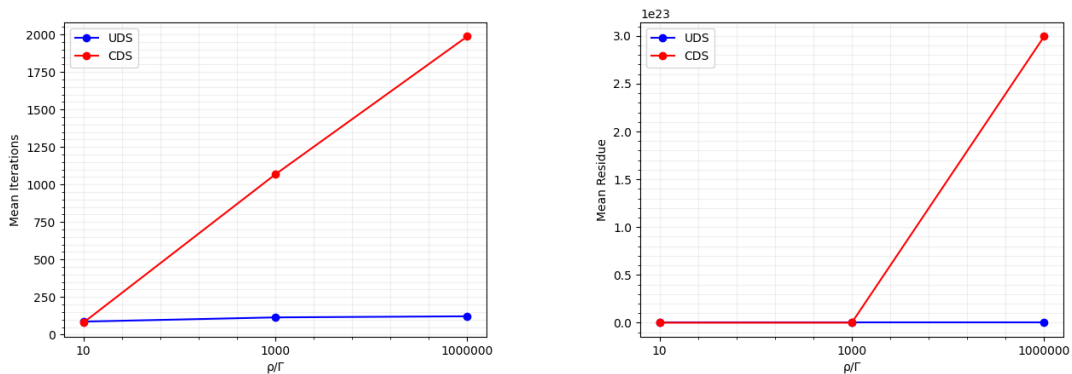


Figure 11: Convective scheme comparison.

2.1.2 Model Validation

As mentioned above, the model will be compared to the provided reference values for the Smith-Hutton problem, these values were obtained using CDS but this report will compare both UDS and CDS. This problem consists of a defined velocity field in a control volume where the presence of a transport variable is introduced. The validation will consist of comparing the steady-state solution at the boundaries with reference values for this particular problem in three main scenarios that consist of changing the ratio between the density ρ and the diffusive transport coefficient Γ . For this test, Γ will remain fixed at 1 while ρ will be evaluated at $10^1, 10^3$, and 10^6 . The results of this test cases are shown in Fig. 13; as it can be seen the, model accurately matched the expected results at the boundaries, with a slight expected deviation for UDS. Con-

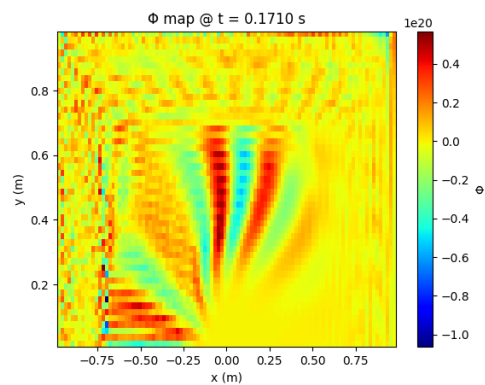


Figure 12: CDS divergence for $\rho/\Gamma = 10^6$

trary to CDS which is second-order accurate, UDS is first-order accurate, this implies certain mathematical simplifications and as such, a slight error against the exact solution. This effect is generally seen in the smoothing of sharp gradients as can be seen in these results. In addition, the obtained steady-state solutions for UDS simulations are shown in Figure 14.

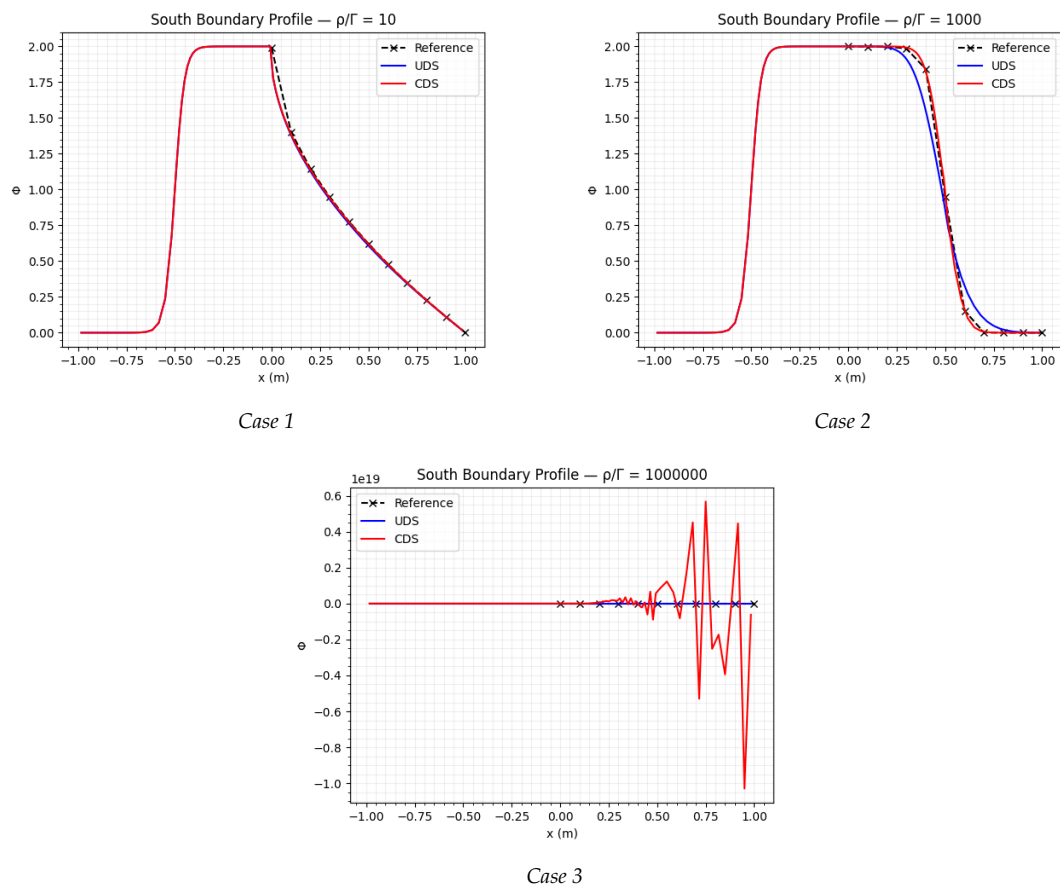


Figure 13: Comparison of model results with results provided in problem statement.

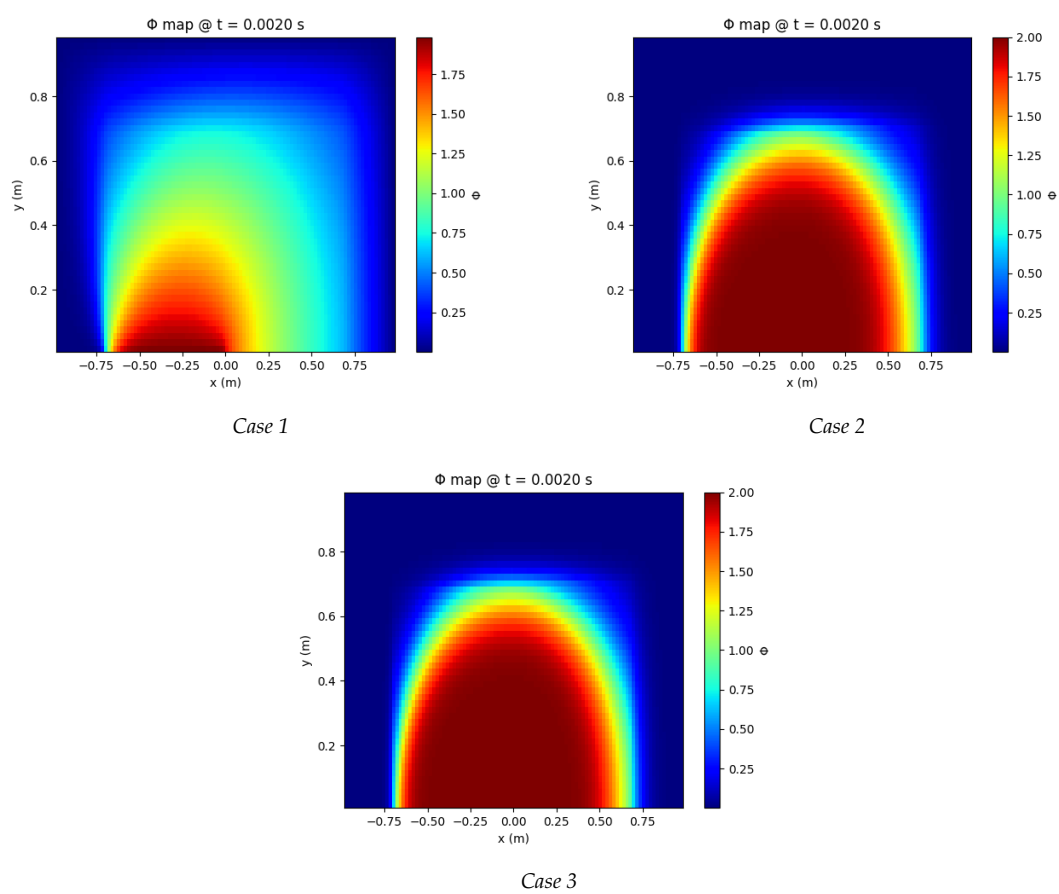


Figure 14: UDS results for all cases.

3 2D Navier-Stokes Model

This chapter will describe the implementation of a numerical solver for 2D Navier-Stokes problems, using the Fractional Step Method and doing the temporal discretization using an Adams-Bashforth method. The Navier-Stokes equations describe the movement of fluids by describing multiple characteristics of the movement in a system of equations. The equations shown in Eq. 25, 26, 27 describe the mass conservation, momentum conservation, and energy conservation respectively. As it can be seen, these equations follow the same structure behind transport equations shown in Chapter , with the corresponding accumulation, convective, diffusive, and source terms. The model will be evaluated against reference values provided for the Lid-Driven Case problem [?], which simplifies the model by removing the energy conservation evaluation. As such, the model will require only 3 equations, one for mass conservation, and the 2 geometric components of the momentum conservation.

$$\nabla \cdot \mathbf{v} = 0 \quad (25)$$

$$\rho \frac{\partial \mathbf{v}}{\partial t} + \rho(\mathbf{v} \cdot \nabla) \mathbf{v} = \mu \nabla^2 \mathbf{v} - \nabla P \quad (26)$$

$$\rho c_p \frac{\partial T}{\partial t} + \rho c_p \nabla(\mathbf{v}T) = \nabla(\lambda \nabla T) + \dot{q}_V \quad (27)$$

As mentioned above, this model uses the Adams-Bashforth discretization method, which relies on two different instants to evaluate the next iteration. Eq. 28 shows the expression used for this discretization method, where $R(\mathbf{v})$ corresponds to the convective and diffusive terms evaluated at the corresponding time-step and is described by Eq. 29. The discretization for this expression will follow the discretization implemented in Chapter 2.

$$\rho \frac{\mathbf{v}^{n+1} - \mathbf{v}^n}{\Delta t} = -\nabla P^{n+1} + \frac{3}{2}R(\mathbf{v}^n) - \frac{1}{2}R(\mathbf{v}^{n-1}) \quad (28)$$

$$R(\mathbf{v}) = \mu \nabla^2 \mathbf{v} - \rho(\mathbf{v} \cdot \nabla) \mathbf{v} \quad (29)$$

Since this is now a coupled solver solving for multiple variables instead of solving for a single one, the configuration file will now require the relevant details and information about every variable. As such, initial values and boundary conditions now need to be specified separately for the main solver variable and for the secondary, explicitly calculated, variables. In addition, a new Field probe was introduced specifically for the velocity field. Table 11 describes all the configuration parameters available for the current model.

3.1 Fractional Step Method

This section will explain the different modifications that were implemented in the model to solve the current test case. As mentioned above, there is a requirement to solve multiple transport variables at the same time that are mutually dependent. Since the velocity and pressure

need to be computed simultaneously, the Fractional Step Method separates this process into three separate steps. The first step consists of evaluating the velocity field while ignoring the pressure field. This calculated velocity field will not be an accurate solution for the next time-step; however, it can be used to evaluate the pressure field at the next time-step. After that second step, the only remaining operation is to correct the calculated velocity field with the influence of the now known pressure field. The three steps can be summarized by the expressions shown in Eqs. 30, 31, 32, showcasing the methodology used to obtain the solution to this particular case. Since the two velocity components are calculated explicitly, there is no need to employ a numerical solver for those variables. On the other hand, the pressure equation will have the same structure as a diffusion equation as it can be seen in Eq. 31.

$$\mathbf{v}^p = \mathbf{v}^n + \frac{\Delta t}{\rho} \left[\frac{3}{2} R(\mathbf{v}^n) - \frac{1}{2} R(\mathbf{v}^{n-1}) \right] \quad (30)$$

$$P_p^{n+1} = \frac{1}{a_p} (a_n P_N^{n+1} + a_s P_S^{n+1} + a_e P_E^{n+1} + a_w P_W^{n+1} - \frac{\rho \Delta S}{\Delta t} (u_e^p - u_w^p + v_n^p - v_s^p)) \quad (31)$$

$$\mathbf{v}_p = \mathbf{v}_p^p - \frac{\Delta t}{\rho} \nabla P^{n+1} \quad (32)$$

3.1.1 Staggered Meshes

One may recall that the discretization model discussed presented earlier in this document requires the evaluation of flow at the boundary of each control volume. As such, rather than computing the pressure and velocity components using a standard mesh, it is beneficial to stagger independent meshes for each one of our variables. By doing this, the computed nodes for each variable are already located at the necessary positions for their interactions, eliminating the need for interpolation during the calculation process. Figure 15 show the distribution of nodes on top of the main variable mesh grid.

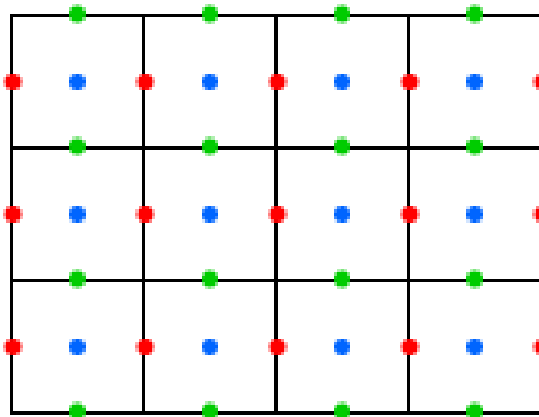


Figure 15: Staggered Mesh representation. Blue nodes correspond to pressure, red nodes to the first velocity component ($u = v_x$), and green nodes to the second velocity component ($v = v_y$).

3.2 Model Validation

As mentioned above, the model will be validated by comparing the results to given reference values for the Lid-Driven Case problem with $Re = 100, 400$ using both CDS and UDS. Re is defined by Eq. 33. No numerical study will be included in this section as the velocity components are calculated explicitly and the pressure equation is solved using the same model described in previous chapters. The reference values given are distributed across the mid-sections of each axis of the domain ($x = 0.5, y = 0.5$), and will be plotted against the model solution for CDS and UDS.

$$Re = \frac{\rho UL}{\mu} \quad (33)$$

Fig. 16 shows the comparison of the model results to the given reference values, demonstrating that the present model was able to replicate the expected behaviour. In addition, the results show a clear difference between the diffusion-dominant $Re = 100$ case, and the convection-dominant $Re = 400$ case. The diffusion-dominant case will have a smoother distribution as the effect of velocity gradients is not too impactful yet. Furthermore, since the effect of the velocity gradient is still small, the results for UDS and CDS will be practically the same. On the other hand, the profiles for the convection-dominant case show clearly defined inflection points and a noticeable separation between the results for UDS and CDS. When the simulation becomes convection-dominant, the effect of the velocity gradients will be more important, and the difference between the advective schemes will be more clear. Figures 17, 18 show the results obtained for the velocity field in steady-state for the simulations, both as a colormap of the magnitude, and as a vector map showing the flow direction.

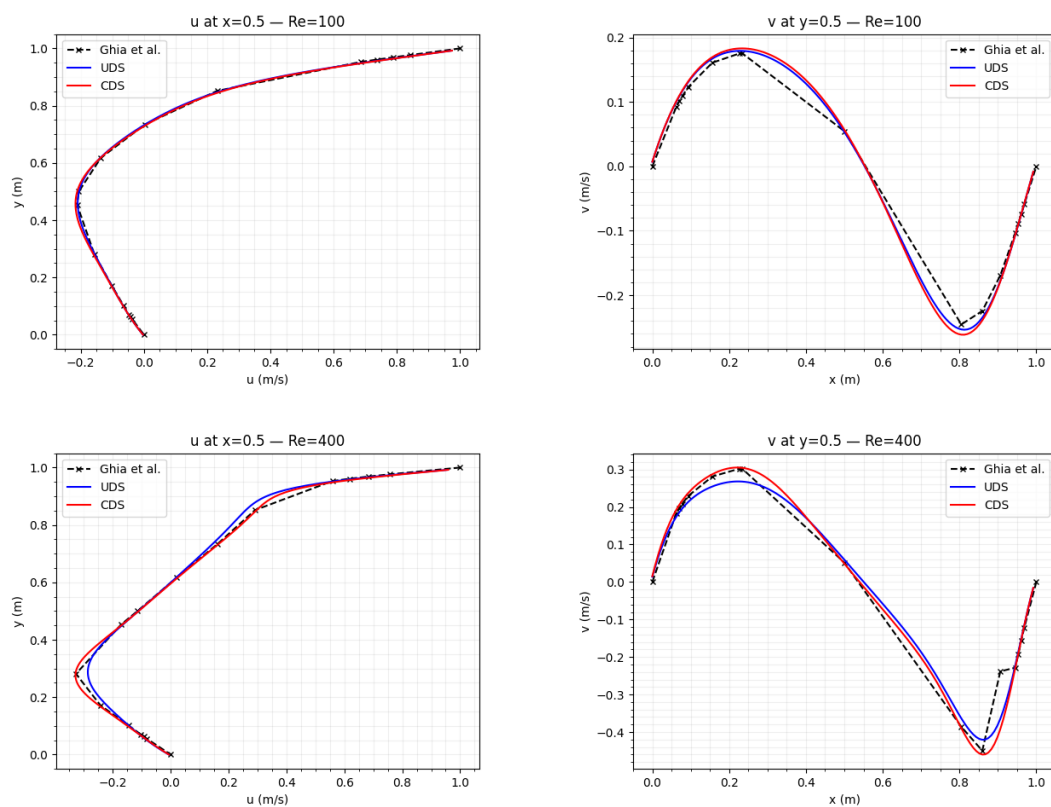


Figure 16: Comparison of model results with results provided in problem statement.

Group	Parameter	Description
Numerical Data	tempScheme spatScheme solver tolNumeric tolTemporal maxIterations endTime timeStep	Temporal scheme selection for simulation. Convective scheme selection for simulation. Solver algorithm for simulation. Tolerance for the numerical solver. Tolerance for the steady-state convergence. Limit of iterations per timestep. Last instant of the simulation. Time increment after each instant.
Physical Data	PHI0 VF0 materials velocityField	Initial value for main solver variable. List with initial value for each component of velocity. List of vectors with thermophysical properties corresponding to each material. List of expressions for each component of the velocity field.
Geometrical Data	width sections	Dimension in the secondary axis. List of vectors indicating the distribution of materials in the simulation. Also includes amount of nodes and internal heat generation.
Mesh Data	strength centering kappa delta meshAlgorithm N refinement	Cubic algebraic mesh parameter. Cubic algebraic mesh parameter. Power law mesh parameter. Hyperbolic tangent mesh parameter. Algorithm for mesh generation. List specifying total number of nodes for each axis. List of vectors specifying refinement configuration for each axis.
Boundary Data	boundaries boundariesVelocity	List of vectors specifying the boundary conditions on each side for the main solver variable. List of vectors specifying the boundary conditions on each side for the velocity components.
Probe Data	probes	List of vectors specifying the probes configured for the simulation.
Medic Data	medicOn	Control argument for diagnostic tool.

Table 11: Configuration file structure.

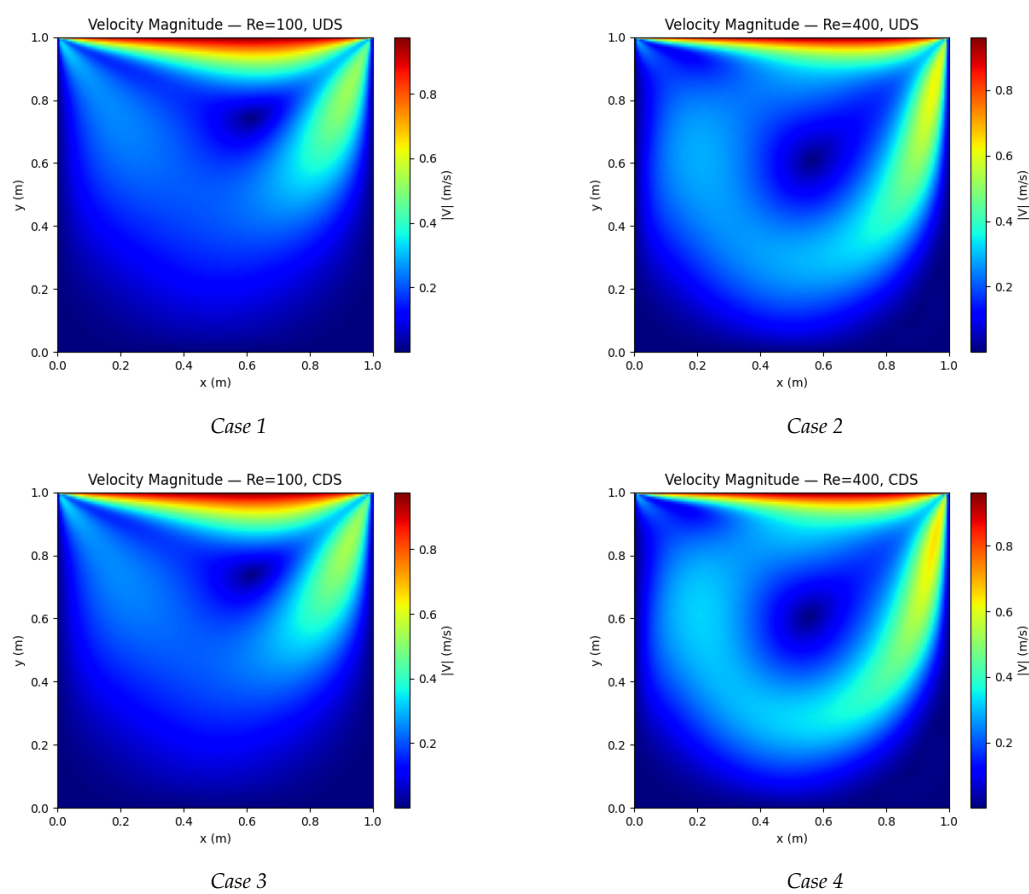


Figure 17: NS velocity profile for all test cases.

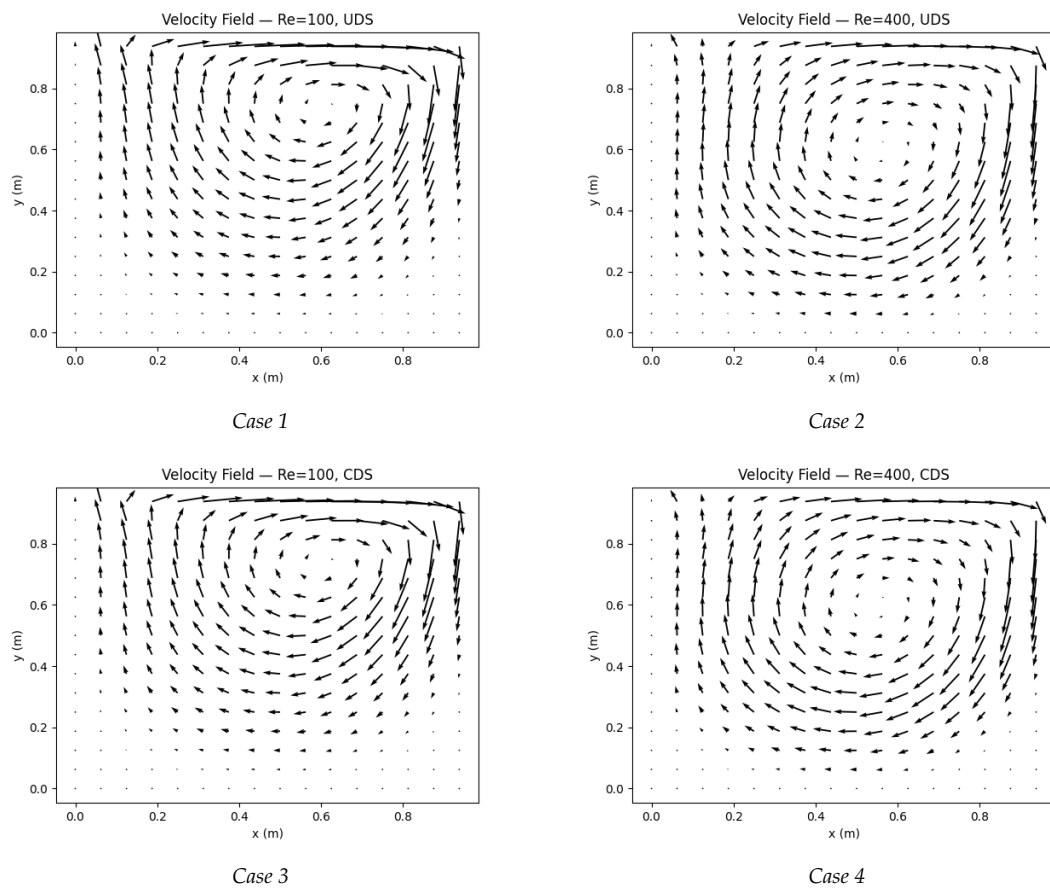


Figure 18: NS velocity vector map for all test cases.

Conclusions

This document has covered the development and evolution of a Heat and Mass Transfer solver across different benchmarks of increasing complexity. Across the different stages of development of this model, different complexities regarding the implementation of numerical methods and the simplification of physical models have been observed and analyzed. In addition, the model developed during this project will serve as a stepping stone towards more complex implementations. There are still several improvements that can be made to the model, particularly in terms of optimization of variables and standardization of certain algorithms. The model will continue to be maintained to expand its capacity and increase its performance.

1. The Pure Diffusion Model served as a first step in defining the architecture of the model. This structure consists of the 5 main components (Material, Mesh, Discretizer, Solver, and Probe) that handle the different operations required for the solver; storing data about the domain, about the physical phenomena, solving the model, and storing the desired results. This system allows for organized and easy-to-read code, which makes it easier to work on long-term maintenance of the model. Furthermore, this model was used as a baseline comparison for the linear solvers implemented in the model (GS and CG).
2. The Convection Diffusion Model served as an introduction to the advective scheme, which determines the numerical approach for approximating the convective effect of flow at the walls. The formulation proposed by Patankar allows for a standardized model that can handle different advective schemes under a single formulation. It is important to highlight the observed behaviour of CDS when used to solve convection-dominant problems, as this model can lose diagonal dominance for $|Pe| > 2$, showing a checkerboarding effect. For high Pe cases on the other hand, UDS can provide a more stable solution while sacrificing precision due to gradient smoothing.
3. The Navier-Stokes Model introduced the Fractional Step Method, which consists of separating the velocity components from the pressure component and solving them discretely by using the Adams-Bashforth discretization method to obtain second order accuracy. Furthermore, Staggered Meshes were used to eliminate the need for interpolating values at the faces; choosing to locate them at the required locations from the start. Finally, the difference between a diffusion-dominant and convection-dominant scenario was also evident in this tests. However, since the velocity is decoupled and calculated explicitly with Adams-Bashforth, the same checkerboarding apparent for the convection-diffusion case was not seen here.
4. All models were successfully validated against the benchmark reference values.

3.3 Future Work

1. During the development of the model, the code architecture has changed significantly. As the model kept expanding the, new variables and structs were introduced to store the required information and better organize the code. While there were several changes in structure when moving from a pure-diffusion model to a convection-diffusion model, the main changes (particularly to the Mesh) were made when moving to the Navier-Stokes model. Since the first two models solved a single transport variable, Mesh.cpp stored all relevant variables directly. On the other hand, the NS solver requires multiple transport variables to be evaluated at the same time; and as such, they require their own set of vari-

ables for that purpose. For this purpose, a cleaner struct implementation was developed that organizes each transport variable independently and stores different metadata to reduce the operations done by the discretizer. The convection-diffusion solver should be modified to operate under the same parameters, modifying the logic to fit the new struct definition and standardizing the model.

2. The current model can still be expanded to support more complex cases, with the next step being solving natural convection problems. This requires the inclusion of the energy conservation equation in the model as well. In addition, the current model can still be extended to support a 3D domain; however, this change implies a larger overhaul of several functions in the model and will be considered for implementation at a later date.
3. Further studies about the nature of the numerical solver can be included to deepen the understanding and implication of this decisions. For instance, only CDS and UDS were tested during the development of this project even though the Hybrid, Exponential, and Power Law schemes are also implemented. With more time and computational resources, additional tests for the 5 different schemes could be done in different edge scenarios.

References

- [1] Suhas V. Patankar. *Numerical Heat Transfer and Fluid Flow*. Series in Computational Methods in Mechanics and Thermal Sciences. Hemisphere Publishing Corporation, Washington, D.C., 1980.